Completeness of Relational Operations in Language Models

Results on a 500-Question Benchmark

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Abstract

We report results for five language models on a 500-question benchmark that asks for *every* item satisfying a condition, over records whose exact answers are obtained by executing SQL. On the 365 questions whose records are in the prompt, mean recall is 0.966–0.996, but only 81–98% of answers are exactly the correct set, and nearly all errors are omitted items. For three models the chance of a complete answer falls significantly as the number of correct items grows. Alternative prompts did not improve completeness. Models that stayed complete used far more reasoning tokens on larger answer sets.

1 Process

Benchmark. The benchmark has 500 questions over 575 synthetic tables (110,146 records) about fictional entities, so answers cannot come from memory. Each question is generated with the SQL query that defines it, and its gold answer is that query's result in SQLite. The 400 *in-context* questions put all records in the prompt; 60 *memory-only* questions filter fixed lists (U.S. states, chemical elements, presidents' surnames); 40 *retrieval* questions supply the top-k BM25 results. For in-context enumeration, exactly $g \in \{5, 10, 20, 50, 100\}$ of $n \in \{50, 200, 500, 1000\}$ rows match, so answer size and table size vary independently. Table 1 lists the slices.

Models. DeepSeek V4 Pro and V4 Flash (OpenCode Go API), Claude Haiku 5.5 (Claude Code CLI 2.1.296, no tools) and MiMo-V2.6 Pro and Flash (OpenRouter), each with its default reasoning, temperature 0 where supported, and a 32,768-token output limit. Answers are returned as JSON.

Metrics. For gold set G and predicted set P: recall $|P \cap G|/|G|$, precision $|P \cap G|/|P|$, and **exact-set accuracy** $\mathbb{1}[P = G]$, the primary measure. Each error is classified as *missing*, *near-miss* (a garbled copy of a missing item), *wrong* (in the data but not qualifying), *made-up* (not in the data) or *duplicate*. An answer *omits* if it misses a correct item and *commits* if it adds an incorrect one.

Runs and statistics. **E1**: all 500 questions, direct prompt. **E2**: a stratified 100-question subset with a "be exhaustive" prompt and a "check every record first" prompt. **E3**: the subset again with the direct prompt. Intervals are 95% (Wilson for proportions, question-level bootstrap for means); model and prompt comparisons use paired question-level differences; trends with answer size use a logistic regression on $\log_2$ of the number of correct items.

Slice	Qs	Variants	Answer size	Context (k tok.)
Enumeration	100	one category filter	5–100	7.6
Enumeration (memory only)	60	U.S. states, elements, presidents	2–50	0.0
Enumeration (retrieval)	40	BM25 top-k, $k \in \{10, 20, 50\}$	5–40	1.1
Selection	60	1–3 conditions	1–92	4.1
Projection	40	DISTINCT; 2–3 attributes	7–39	2.9
JOIN	60	1:1, 1:N, M:N	3–55	3.4
Aggregation	50	COUNT, SUM, MIN/MAX/AVG, GROUP BY	3–367	2.8
Nested query	40	IN, NOT IN, correlated, scalar	3–32	2.7
Set operation	50	UNION, INTERSECT, EXCEPT, 3-way	4–97	0.7

Table 1: Benchmark composition. Answer size is the number of correct items (for aggregation, records aggregated); context is the median data size in the prompt.

2 Results

2.1 Overall completeness

Table 2 shows the 365 in-context set-valued questions. Recall and precision are close to 1 for every model, but exact-set accuracy ranges from 98% to 81%. The two DeepSeek models are tied; the other three are significantly below the best model, by 7 to 17 percentage points.

Model	Exact-set acc. (%)	Δ vs. best (pp)	Mean recall	Mean prec.	Omit. (%)	Comm. (%)
DeepSeek V4 Pro	98 [96, 99]	−0.3 [-2.5, +1.6]	0.996 [0.991, 1.000]	0.997	2	0
DeepSeek V4 Flash	98 [96, 99]	(reference)	0.996 [0.989, 1.000]	0.996	2	1
Claude Haiku 5.5	91 [88, 93]	−7.1 [-10.1, -4.1]	0.995 [0.993, 0.997]	0.999	9	1
MiMo-V2.6 Pro	90 [86, 93]	−8.2 [-11.8, -4.9]	0.987 [0.978, 0.994]	0.992	7	4
MiMo-V2.6 Flash	81 [77, 85]	−16.7 [-20.8, -12.3]	0.966 [0.952, 0.979]	0.970	15	10

Table 2: In-context set-valued questions (E1, $n = 365$ per model), with 95% intervals. Δ: paired difference in exact-set accuracy from the best model. *Omit./Comm.*: share of answers that omit a correct item / add an incorrect or duplicate one.

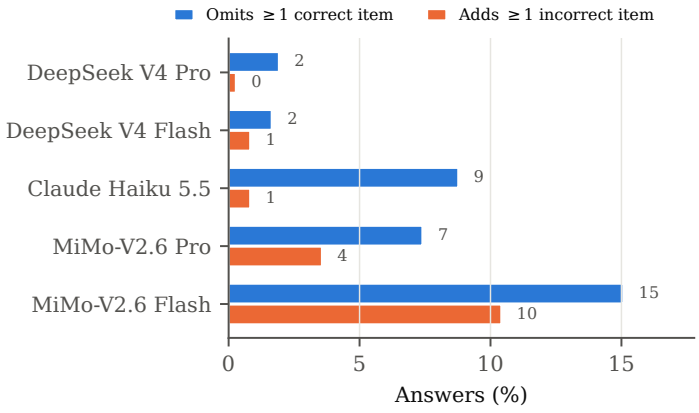


Figure 1: Share of in-context answers that omit at least one correct item, and that add at least one incorrect item (E1).

Errors are almost all omissions (Figure 1; counts in Table 11). Four models together returned 25 wrong items and 0 made-up items across all their answers. MiMo-V2.6 Flash is the exception: it returned 1,032 made-up items, 991 of them on a single 100-item question.

Takeaway: high recall does not mean complete answers; models leave correct items out far more often than they add wrong ones, so incompleteness is not visible in what they list.

2.2 Completeness and answer size

On in-context enumeration (Figure 2, Table 3), every model is complete on at least 93% of questions with up to 10 correct items. With 50 or more, Claude Haiku 5.5 falls to 53%, MiMo-V2.6 Pro to 75% and MiMo-V2.6 Flash to 34%. The decline is significant for DeepSeek V4 Pro, Claude Haiku 5.5 and MiMo-V2.6 Flash, whose odds of a complete answer fall to 0.31–0.41× per doubling of the answer set. DeepSeek V4 Flash fails too rarely to show a trend.

Takeaway: the more items a correct answer contains, the less likely it is to be complete.

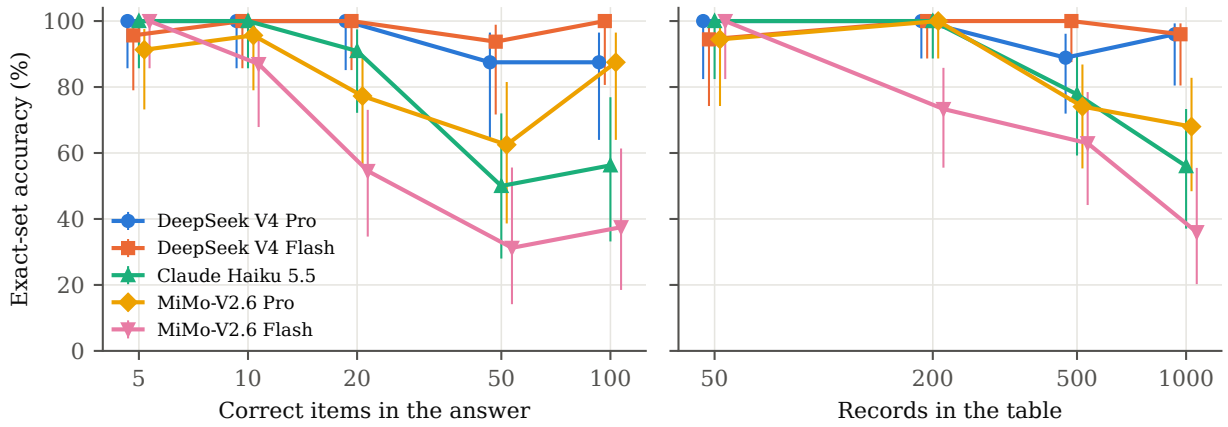


Figure 2: In-context enumeration (E1): exact-set accuracy by number of correct items (left) and table size (right), with 95% Wilson intervals.

Model	≤10 items (%)	≥50 items (%)	Odds ratio per doubling [95% CI]
DeepSeek V4 Pro	100	88	0.31 [0.05, 0.54]
DeepSeek V4 Flash	98	97	not estimable (2 failures)
Claude Haiku 5.5	100	53	0.32 [0.17, 0.45]
MiMo-V2.6 Pro	93	75	0.74 [0.51, 1.05]
MiMo-V2.6 Flash	93	34	0.41 [0.26, 0.55]

Table 3: In-context enumeration (E1): exact-set accuracy for small and large answer sets, and the odds ratio of a complete answer per doubling of the number of correct items (below 1 means degradation).

2.3 By operation

Every operation except set operations produced incomplete answers from at least three models (Table 4). Enumeration, which has the largest answer sets, is hardest; set operations are solved almost perfectly.

Takeaway: the effect is not specific to one operation.

Operation	Qs	DeepSeek V4 Pro	DeepSeek V4 Flash	Claude Haiku 5.5	MiMo-V2.6 Pro	MiMo-V2.6 Flash
Enumeration	100	96	98	83	84	66
Selection	60	97	98	95	93	88
Projection	40	100	95	90	90	82
JOIN	60	100	98	93	88	82
Aggregation	15	93	93	93	87	67
Nested query	40	98	100	90	92	90
Set operation	50	100	100	100	98	100

Table 4: Exact-set accuracy (%) by operation, in-context set-valued questions (E1). Aggregation here is GROUP BY; scalar aggregates are in Table 6.

2.4 Memory-only and retrieval settings

From memory, both DeepSeek models were always exact; the other three missed members of short fixed lists (Table 5), e.g. Claude Haiku 5.5 named Kentucky but not New Jersey as a state ending in "y". In the retrieval setting every model returned every correct record that was retrieved, so recall (0.869 for all) equals the retriever's recall.

Takeaway: the retrieval slice, as built, measures the retriever rather than the models.

Model	Memory: exact (%)	Memory: recall	Retrieval: recall	Retrieval: recall over retrieved
DeepSeek V4 Pro	100	1.000	0.869	1.000
DeepSeek V4 Flash	100	1.000	0.869	1.000
Claude Haiku 5.5	87	0.952	0.869	1.000
MiMo-V2.6 Pro	83	0.955	0.869	1.000
MiMo-V2.6 Flash	83	0.958	0.869	1.000

Table 5: Memory-only and retrieval settings (E1).

2.5 Scalar aggregation

On 35 COUNT, SUM, MIN, MAX and AVG questions (Table 6), both DeepSeek models were always exact. Of the other models' 11 errors, 9 were undercounts.

Takeaway: wrong totals are mostly too low, consistent with dropped items.

Model	Qs	Exact (%)	Under-counts	Over-counts
DeepSeek V4 Pro	35	100	0	0
DeepSeek V4 Flash	35	100	0	0
Claude Haiku 5.5	35	94	2	0
MiMo-V2.6 Pro	35	91	3	0
MiMo-V2.6 Flash	35	83	4	2

Table 6: Scalar aggregation questions (E1).

2.6 Consistency across runs

Two independent samples of the subset agreed on 87–100% of verdicts (Table 7). Of 13 failures re-run three times each (Table 8), some recurred every time: Claude Haiku 5.5's missing U.S. states, and DeepSeek V4 Pro's and MiMo-V2.6 Flash's misses on the same 50-item list. When MiMo-V2.6 Pro failed on that list it omitted the same three employees each time. Others, including Claude Haiku 5.5's misses on selection, JOIN, projection and COUNT questions, never recurred.

Takeaway: some failures are systematic, but many are run-to-run variation, so completeness is best reported as a rate over many questions.

Model	Qs	Same verdict (%)	Questions that flipped
DeepSeek V4 Pro	87	98 [92, 99]	2
DeepSeek V4 Flash	93	100 [96, 100]	0
Claude Haiku 5.5	93	94 [87, 97]	6
MiMo-V2.6 Pro	93	91 [84, 96]	8
MiMo-V2.6 Flash	93	87 [79, 92]	12

Table 7: Agreement between two independent samples of the direct prompt (E1 vs. E3), with 95% Wilson intervals.

Model	Question	Answers	Records	Main-run outcome	Reproduced
Claude Haiku 5.5	`enum-par-013`	2	50	recall 0.50, 1 missing	2/3
Claude Haiku 5.5	`enum-par-021`	11	50	recall 0.73, 3 missing	3/3
Claude Haiku 5.5	`sel-029`	13	400	recall 0.85, 2 missing	0/3
Claude Haiku 5.5	`join-004`	55	100	recall 0.93, 4 missing	0/3
Claude Haiku 5.5	`nest-022`	13	200	recall 0.92, 1 missing	2/3
Claude Haiku 5.5	`proj-031`	37	300	recall 0.89, 4 missing	0/3
Claude Haiku 5.5	`agg-015`	12	400	answered 11 (gold 12)	0/3
Claude Haiku 5.5	`enum-ctx-029`	50	500	recall 0.96, 2 missing	0/3
DeepSeek V4 Pro	`enum-ctx-029`	50	500	recall 0.98, 1 missing	3/3
MiMo-V2.6 Flash	`enum-ctx-029`	50	500	recall 0.94, 3 missing, 3 wrong	2/3*
MiMo-V2.6 Pro	`enum-ctx-029`	50	500	recall 0.92, 4 missing	1/3
MiMo-V2.6 Flash	`enum-ctx-035`	100	1000	recall 0.52, 46 missing, 2 near-miss, 19 wrong, 371 made-up, 34 duplicate	2/3
DeepSeek V4 Flash	`enum-ctx-018`	5	50	recall 0.80, 1 near-miss	0/3

Table 8: Selected failures, each re-run three more times on the same model. *Reproduced*: re-runs again not exactly correct (* marks a re-run with no parseable answer, not counted). MiMo cases come from an earlier OpenCode Go run of the same models; DeepSeek and MiMo re-runs used OpenRouter.

2.7 Prompt variants

Neither the "be exhaustive" nor the "check every record" prompt changed exact-set accuracy significantly for any model; every paired interval includes zero (Table 9).

Takeaway: simple prompting does not fix incompleteness.

Model	Qs	Exhaustive − direct (pp)	Qs	Row-by-row − direct (pp)
DeepSeek V4 Pro	–	–	–	–
DeepSeek V4 Flash	93	−2.2 [-5.4, +0.0]	93	+0.0 [+0.0, +0.0]
Claude Haiku 5.5	93	+1.1 [-3.2, +6.5]	93	−2.2 [-6.5, +2.2]
MiMo-V2.6 Pro	93	+1.1 [-4.3, +6.5]	93	+4.3 [-1.1, +9.7]
MiMo-V2.6 Flash	93	+0.0 [-8.6, +9.7]	93	+6.5 [-2.2, +15.1]

Table 9: Paired change in exact-set accuracy (percentage points) from the direct prompt on the subset (E2), with 95% bootstrap intervals. DeepSeek V4 Pro's E2 runs did not finish within its provider's usage limits.

2.8 Reasoning effort

Median reasoning length rose with answer size for the models that stayed complete (Figure 3): from 1,050 to 8,832 tokens for DeepSeek V4 Pro and from 173 to 6,442 for Claude Haiku 5.5. MiMo-V2.6 Flash often answered after one sentence of reasoning; its 61 answers with under 100 reasoning tokens were complete 62% of the time, against 72% for the rest.

Takeaway: completeness on large answer sets coincides with much longer reasoning; this is an association, not a controlled result.

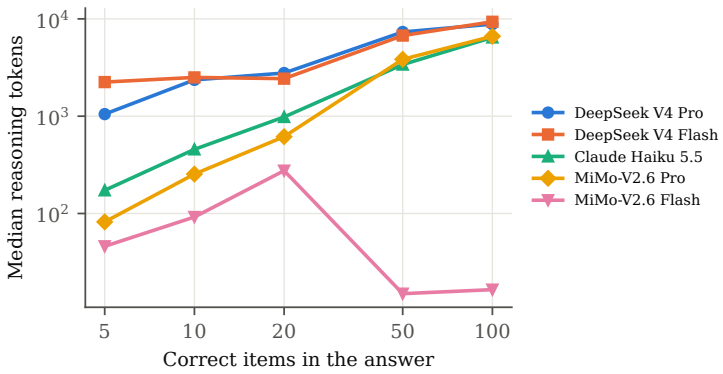


Figure 3: Median reasoning tokens on in-context enumeration by number of correct items (E1).

2.9 Usage

Across all runs the five models made 4,827 calls; 77–97% of output tokens were reasoning (Table 10).

Model	Access	Calls	Input (M)	Output (M)	Reasoning (%)	Cost (USD)	Latency (s)
DeepSeek V4 Pro	OpenCode Go	1,027	3.83	2.21	97	6.87	23.4
DeepSeek V4 Flash	OpenCode Go	1,307	5.08	3.02	93	2.52	12.5
Claude Haiku 5.5	Claude Code CLI	893	8.04	1.88	87	2.49	7.7
MiMo-V2.6 Pro	OpenRouter	800	5.77	2.00	89	3.29	16.6
MiMo-V2.6 Flash	OpenRouter	800	5.78	1.72	77	1.25	13.1

Table 10: Usage across all experiments, including retried calls. Input and output in millions of tokens; *Reasoning*: share of output tokens; cost at provider list prices (Claude Haiku 5.5 and the OpenCode Go models were used through subscriptions); latency is the median per call.

3 Summary

- Mean recall of 0.966–0.996 corresponds to exact-set accuracy of only 81–98%; 2–15% of in-context answers omit at least one correct item.
- Errors are omissions, not additions, for four of five models; one model fabricates items on long lists.
- Completeness falls with answer size, significantly for three models.
- Prompt variants do not help; complete models reason far longer on large answer sets.
- Caveats: synthetic data from one seed; one sample per question in E1 (0–13% of verdicts change on a second sample); the retrieval slice measures the retriever; DeepSeek V4 Pro lacks E2 results; a stopped Claude Haiku 4.5 run (90 questions) is excluded.

Code, data, raw responses and run manifests:
https://github.com/daryl-888/llm-relational-completeness.
Statistics follow E. Miller, "Adding Error Bars to Evals" (2024), https://arxiv.org/abs/2411.00640.

A Additional results

Model	Missing	Near-miss	Wrong	Made-up	Duplicate
DeepSeek V4 Pro	16	0	1	0	0
DeepSeek V4 Flash	43	1	3	0	0
Claude Haiku 5.5	75	0	6	0	0
MiMo-V2.6 Pro	104	0	15	0	0
MiMo-V2.6 Flash	492	7	311	1,032	316

Table 11: Error counts by type on in-context set-valued questions (E1).

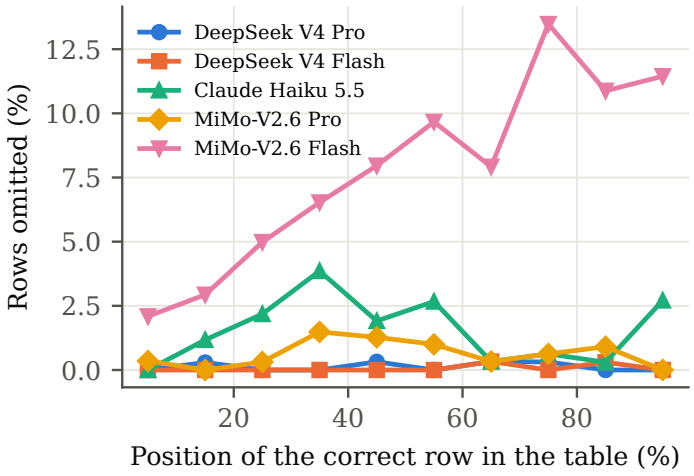


Figure 4: Share of correct rows omitted by the row's position in the table (in-context enumeration, E1). MiMo-V2.6 Flash omits rows near the end more often; the other models omit too few rows for a clear pattern.